1) $L(\{x_n, t_n\}^N, \theta) = \frac{1}{2N} \|X\theta - t\|_2^2$

$$\nabla_\theta L(\cdot) = \nabla_\theta \frac{1}{2N} (X\theta - t)^T (X\theta - t)$$

$$= \nabla_\theta \frac{1}{2N} [\theta^T X^T X\theta - \theta^T X^T t - t^T X\theta + t^2]$$

$$= \nabla_\theta \frac{1}{2N} [\theta^T X^T X\theta - 2\theta^T X^T t + t^2] \quad , \quad \theta^T X^T t \text{ is scalar so } = tX\theta$$

$$= \frac{1}{2N} [(X^T X + (X^T X)^T)\theta - 2X^T t]$$

$$= \frac{1}{2N} [2X^T X\theta - 2X^T t] \quad , \quad X^T X \text{ symmetric so } (X^T X)^T = X^T X$$

$$\Rightarrow \frac{1}{2N} [2X^T X\theta - 2X^T t] = 0$$

$$\rightarrow X^T X\theta - X^T t = 0$$

$$\Rightarrow \hat{\theta} = (X^T X)^{-1} X^T t$$

2.

$$\frac{1}{2N}\left\| \begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}\theta_1\\ \theta_2\end{bmatrix} - \begin{bmatrix}1\end{bmatrix}\right\|_2^2 = \frac{1}{2N}\left(\begin{bmatrix}1, & 2\end{bmatrix}\begin{bmatrix}\theta_1\\ \theta_2\end{bmatrix} - 1\right)^2$$

$$= \frac{1}{2N}\left(\theta_1 + 2\theta_2 - 1\right)^2$$

$$\frac{\partial}{\partial\theta_1} = \frac{1}{N}\left(\theta_1 + 2\theta_2 - 1\right) = 0 \qquad \frac{\partial}{\partial\theta_2} = \frac{1}{N}\left(\theta_1 + 2\theta_2 - 1\right)2 = 0$$

$$\theta_1 + 2\theta_2 - 1 = 2\theta_1 + 4\theta_2 - 2$$

$$0 = \theta_1 + 2\theta_2 - 1$$

3. a) Let $t = 0$

then $\theta_1 = \theta_0 - \eta \nabla_\theta L(D_0; \theta_0)$

$= 0 - \eta \quad [-x_0 y_0]$

$= \eta y_0 x_0$

$= \alpha_0 x_0 \qquad , \alpha_0 = \eta y_0$

$L(D_1; \theta_1) = \frac{1}{2}(x_1 \theta_1 - y_1)^2$

$= \frac{1}{2}\left[2x_1^T x_1 \theta_1 - 2x_1^T y_1\right]$

$= \left(x_1^T x_1 \theta_1 - x_1^T y_1\right)$

let $t \geq 0$

Suppose $\theta_t = \sum_{i=1}^{t-1} \alpha_i x_i$, are the points sampled so far

then $\theta_{t+1} = \theta_t - \eta \nabla_\theta L(D_t; \theta_t)$

$= \theta_t - \eta\left[\underline{x_t^T x_t \theta_t} - y_t x_t\right]$

$= \theta_t - \eta x_t^T x_t \theta_t + \eta y_t x_t$

$= \left[1 - \eta x_t^T x_t\right] \theta_t$

$= \beta \theta_t + \alpha_t x_t \quad \Rightarrow$ update $\alpha_i' = \beta \alpha_i$

$\Rightarrow = \sum \alpha_i' x_i + \alpha_t x_t$

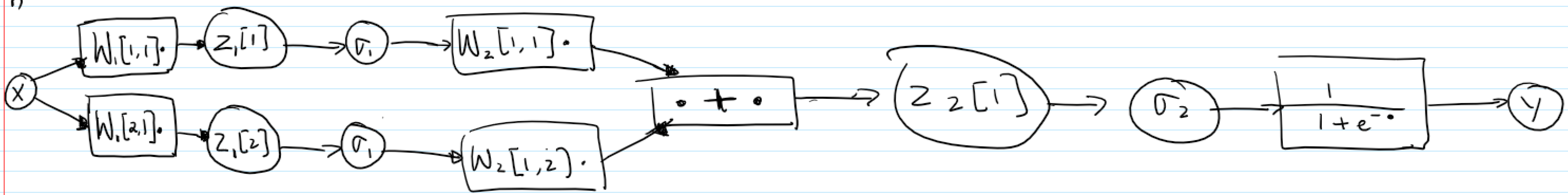
$= \sum_{i=1}^{t} \alpha_i x_i$

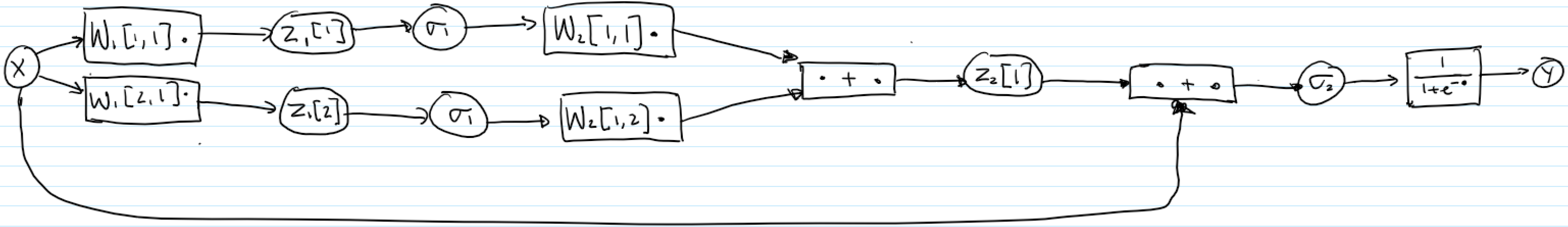
b) $X\theta = t \Rightarrow XX^T\alpha = t$ from a)

$\Rightarrow \alpha = (XX^T)^{-1}t$

$\theta_{SGD} = X^T\alpha \Rightarrow \theta_{SGD} = X^T(XX^T)^{-1}t$

2. 1)


X
W1[1,1]·
W1[2,1]·
Z1[1]
Z1[2]
σ1
σ1
W2[1,1]·
W2[1,2]·
· + ·
Z2[1]
σ2
1/(1+e^-·)
Y


X
W1[1,1]·
W1[2,1]·
Z1[1]
Z1[2]
σ1
σ1
W2[1,1]·
W2[1,2]·
· + ·
Z2[1]
· + ·
σ2
1/(1+e^-·)
Y

2. for f): $$\frac{\partial \ell(y,t)}{\partial x} = \frac{\partial \ell}{\partial y} \frac{\partial y}{\partial \sigma_2} \frac{\partial \sigma_2}{\partial z_2[1]} \left[\frac{\partial z_2[1]}{\partial \sigma_1} \frac{\partial \sigma_1}{\partial z_1[1]} \frac{\partial z_1[1]}{\partial x} + \frac{\partial z_2[1]}{\partial \sigma_1} \frac{\partial \sigma_1}{\partial z_1[2]} \frac{\partial z_1[2]}{\partial x} \right]$$

for g): $$\frac{\partial \ell(y,t)}{\partial x} = \frac{\partial \ell}{\partial y} \frac{\partial y}{\partial \sigma_2} \left[\frac{\partial \sigma_2}{\partial x} + \frac{\partial \sigma_2}{z_2[1]} \left[\frac{\partial z_2[1]}{\partial \sigma_1} \frac{\partial \sigma_1}{\partial z_1[1]} \frac{\partial z_1[1]}{\partial x} + \frac{\partial z_2[1]}{\partial \sigma_1} \frac{\partial \sigma_1}{\partial z_1[2]} \frac{\partial z_1[2]}{\partial x} \right]\right]$$

3. $$\sigma_1(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \qquad \sigma_1'(x) = 0 \qquad \sigma_2(x) = \max(0, x) \qquad \sigma_2'(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

for f: $$\frac{\partial \ell}{\partial x} = 0$$ trivially

for g: $$\frac{\partial \ell}{\partial x} = \frac{\partial \ell}{\partial y} \frac{\partial y}{\partial \sigma_2} \frac{\partial \sigma_2}{\partial x}$$

$$= \frac{1}{N}(t-y) \frac{e^{-\sigma_2}}{(1+e^{-\sigma_2})^2} \, 1$$ Which doesn't vanish